

▣ 21 p values

Previously we described a statistical inference which allowed us to quantify the probability of an error in making a particular decision. Specifically, the probability of rejecting H_0 when it is true.

E.g. If we reported in a study that the pass rate for learner car drivers in Waterford was different to the rate in Cork, we know that there is a 5% chance that we might be wrong. (If we report that there is no difference between the rates, we have no measure of the probability that we might be wrong.)

We might describe this as the error-based approach – we know the probability of an error (of type I)

An alternative to this is which we might call the evidence-based approach, using what are called p values.

To illustrate, consider again our earlier problem comparing rents paid by male and female first year students.

Male	92.18	94.52	83.71	80.45	98.62	93.58	83.75	87.57	94.5	100.65	89.5	95.56	77.45
Female	108.52	117.12	100.94	102.25	111.42	110.13	101.09	111.03	118.03				

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> t.test(male, female)
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Welch Two Sample t-test
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data: male and female
t = -6.4368, df = 18.468, p-value = 4.123e-06
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -24.91290 -12.66881
sample estimates:
mean of x mean of y
 90.15692 108.94778
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The confidence interval-estimate for the difference between average rent paid for males and females excluded zero so we were able to conclude that there was a statistically significant difference between the two groups.

Note that part of the output also includes a p value = $4.123e-6 = 0.000004123$

We could decide to reject H_0 on the basis of the value of p – specifically if it is less than 0.05

H_0 is rejected $\Leftrightarrow$ the confidence interval estimate excludes zero $\Leftrightarrow p < 0.05$

- This number $4.123e-6$ is called the p value
- The p value is a measure of evidence against the null hypothesis being true.
- It is a probability measure so it must lie in the range $[0, 1]$
- Closer values to zero correspond to stronger evidence that the null hypothesis is false
- Values further from zero correspond to weaker evidence that the null hypothesis is false
- The convention used is to set 0.05 as a threshold. If $p < 0.05$, the null is rejected, and we report a statistically significant finding. If $p \geq 0.05$, we have insufficient evidence of a finding.
- If $p < 0.01$, some report a highly statistically significant finding

Misinterpreting the p values

The p value is a measure of evidence against the null hypothesis but should not be interpreted as an error rate.

For example, suppose a t-test comparing the mean exam score between applied and forensics computing students in an exam yields a p value of 0.035.

Then it may be concluded that there is a statistically significant difference between the two groups.

However, the probability of making an error in drawing such a conclusion is not 3.5%. We can use p values and give them an error interpretation but the error probability is whatever significance level we selected before carrying out the investigation – typically 5%

Neither is it correct to interpret the p value as a probability associated with replication – i.e. if we were to repeat the study with a new sample it is possible that we would fail to conclude that there was a difference between the two groups the second time around. But the probability of this is not 3.5%

The two misinterpretations of p values above are commonly made.

What the p value actually is by definition, is the probability of observing the sample data or more extreme data if the null hypothesis were true

For example, suppose in our problem we found that the difference in exam score between the two groups was 8.6 marks. If the null hypothesis were true, this difference would be zero. Therefore, any difference bigger than this value would be more extreme if the null were true.

Thus, the probability that the actual difference between the two groups being 8.6 mark or more is the p value – in this case 3.5%

Error or evidence approach?

Our first few examples used the error-based approach and involved citing confidence intervals with associated error probabilities. Here we consider the evidence-based approach, which typically involves the examination of p values.

The latter approach is perhaps more commonly used though some criticize the reporting of p values but giving no indication of effect size. For example, it is plainly more informative to report that “exam scores for the applied group exceed those of the forensics group by a statistically significant 8.6 ± 3.5 marks (95% confidence interval)” than it is to say ““exam scores for the applied group exceed those of the forensics group by a statistically significant amount $p < 0.05$ ”

In some cases (e.g. ANOVA that we will see next) it is more usual to take the “p value” approach.